

Foot

F1

Foot

F2

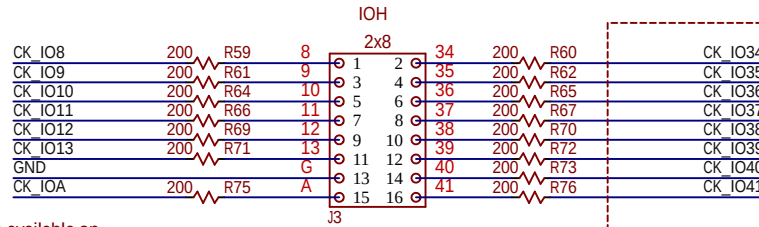
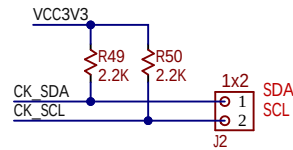
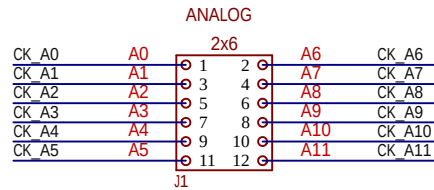
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E3

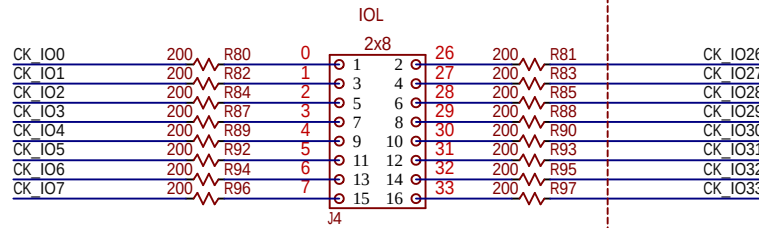
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F4

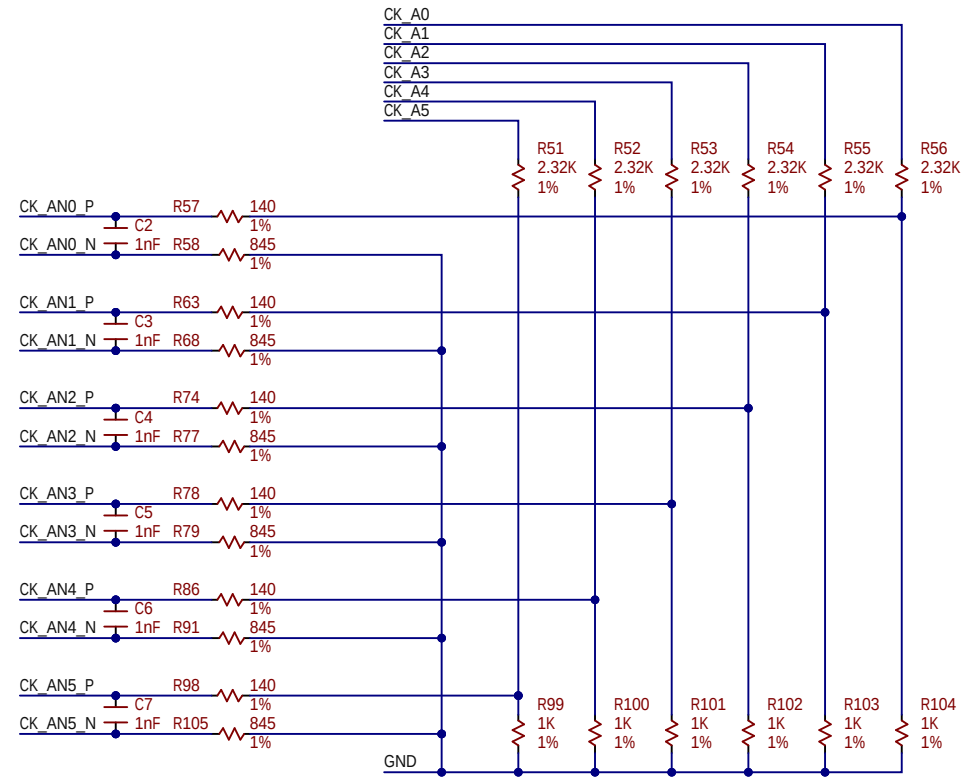
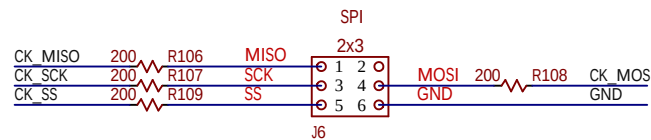
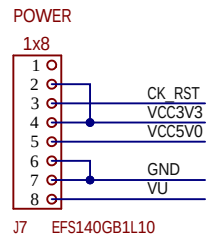
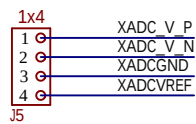
Title		Rev
Arty Z7		D.0
Circuit		
PMODs, User I/O		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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CK_IOA available on
-20 variant only



Available on -20 variant only



Title		Rev
Arty Z7		D.0
Circuit		
chipKIT IO		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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1

2

3

4

A

A

B

B

C

C

D

D

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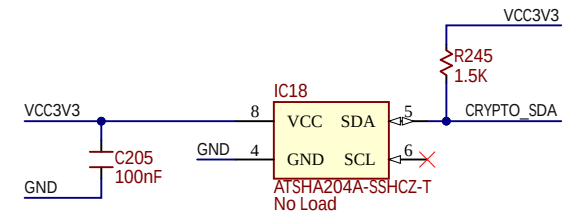
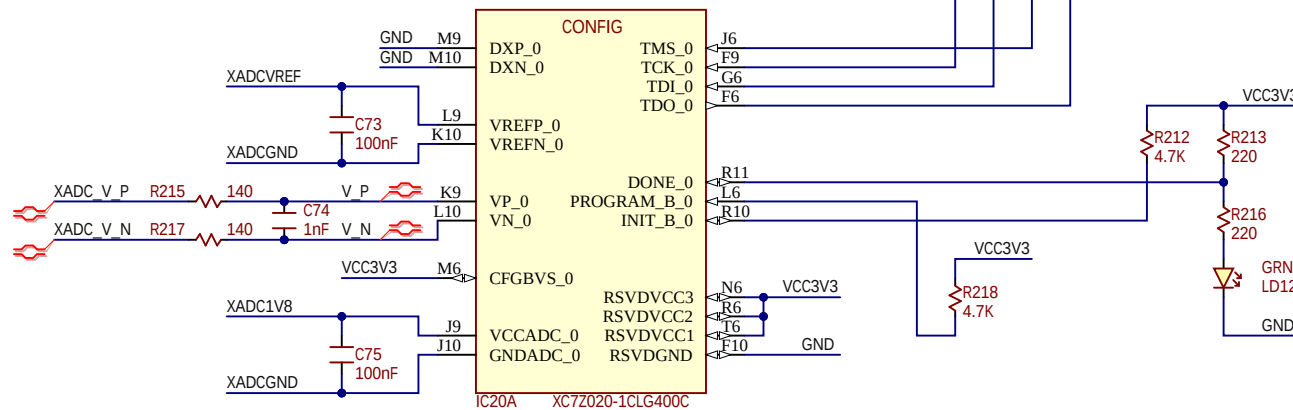
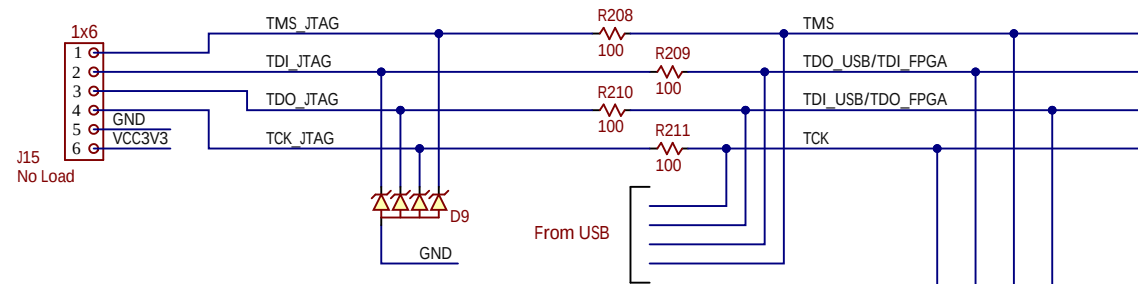
Title		Rev
Arty Z7		D.0
Circuit		
USB PROG, UART		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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1

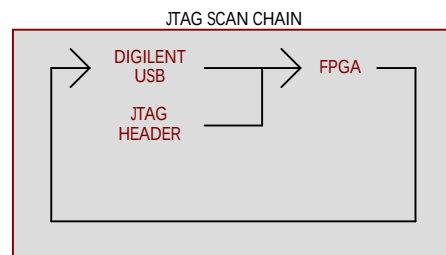
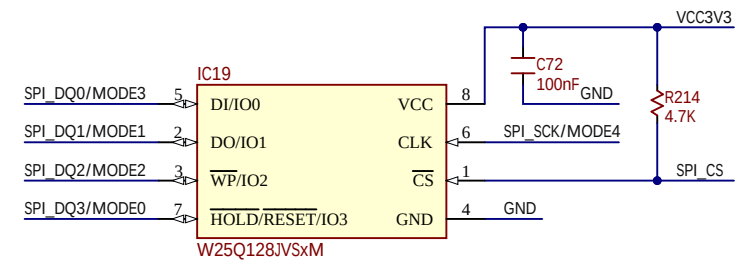
2

3

4



IC19: W25Q128JVSxM (SOIC8) and W25Q128JVPxM (WSON8, 6x5mm) part numbers can be used alternatively, as the PCB footprint works for both.



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Circuit		
FPGA Configuration, Flash		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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A

A

B

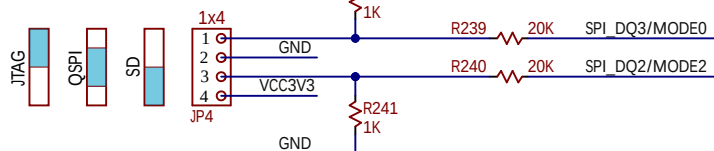
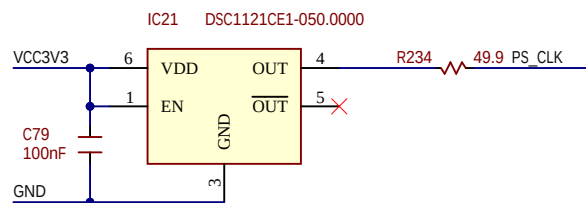
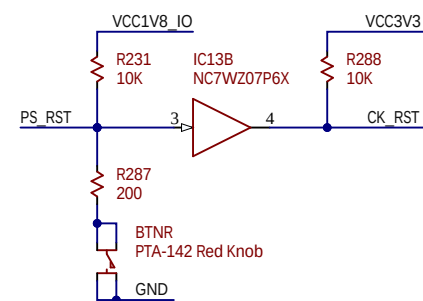
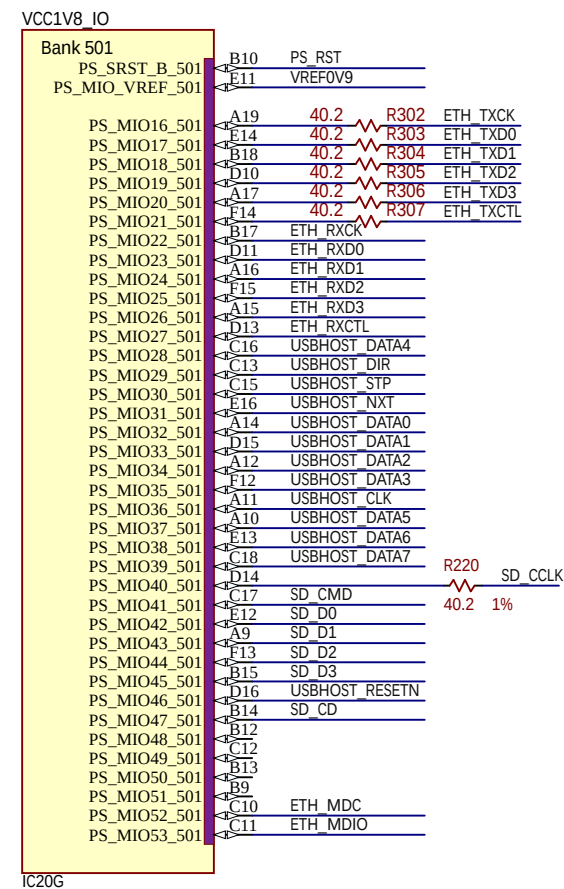
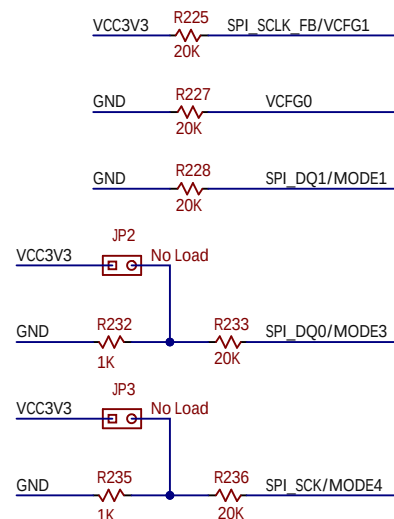
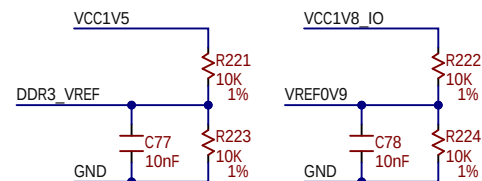
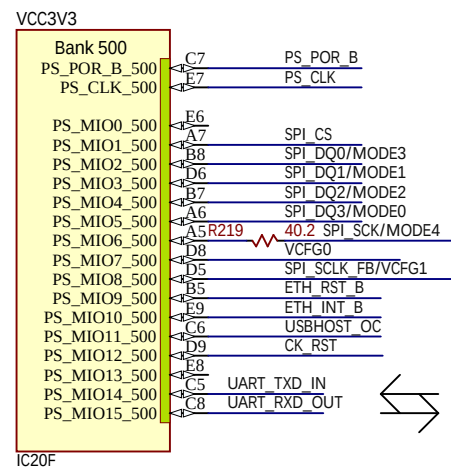
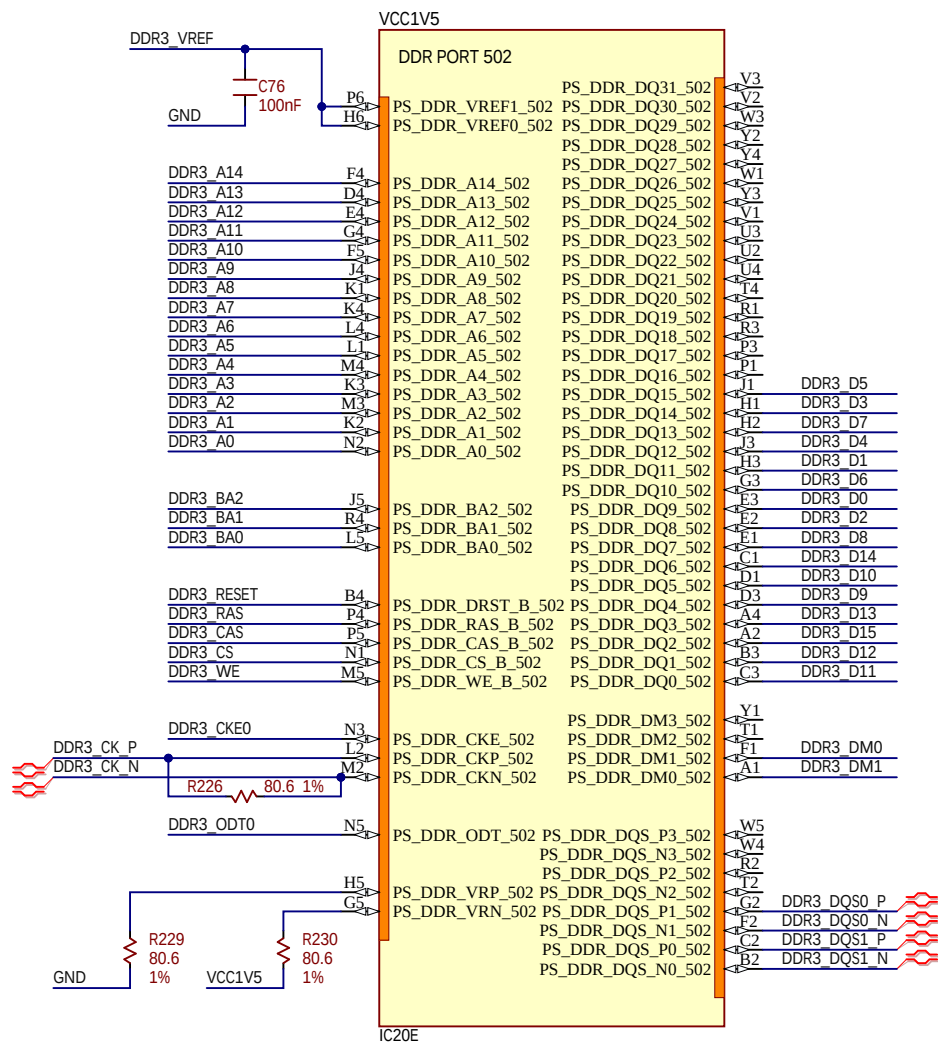
B

C

C

D

D



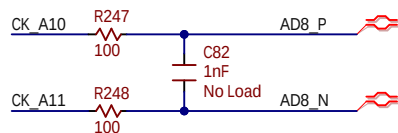
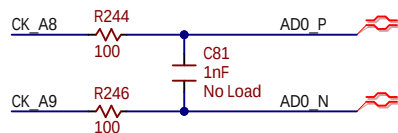
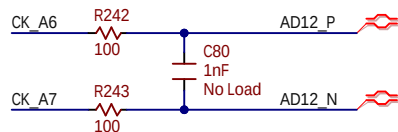
Title		Rev
Arty Z7		D.0
Circuit	DDR and MIO Banks	
Doc#	500-346	
Engineer	IC	
Author	GMA	
Date	6/29/2022	
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A

BANK 13		
IO_L6N_T0_VREF_13	V5	CK IO27
IO_L11P_T1_SRCC_13	U7	CK IO29
IO_L11N_T1_SRCC_13	V7	CK IO30
IO_L12P_T1_MRCC_13	U9	
IO_L12N_T1_MRCC_13	U10	CK A5
IO_L13P_T2_MRCC_13	Y7	CK IO37
IO_L13N_T2_MRCC_13	Y6	CK IO36
IO_L14P_T2_SRCC_13	Y9	CK IO41
IO_L14N_T2_SRCC_13	Y8	CK IO39
IO_L15P_T2_DQS_13	V8	CK IO32
IO_L15N_T2_DQS_13	W8	CK IO38
IO_L16P_T2_13	W10	CK IO34
IO_L16N_T2_13	W9	CK IO40
IO_L17P_T2_13	U9	
IO_L17N_T2_13	U8	CK IO31
IO_L18P_T2_13	W11	CK A2
IO_L18N_T2_13	Y11	CK A0
IO_L19P_T3_13	U5	CK A4
IO_L19N_T3_VREF_13	U5	CK IO26
IO_L20P_T3_13	Y12	CK A1
IO_L20N_T3_13	Y13	CK IOA
IO_L21P_T3_DQS_13	V11	CK A3
IO_L21N_T3_DQS_13	Y10	CK IO33
IO_L22P_T3_13	V6	CK IO28
IO_L22N_T3_13	W6	CK IO35

IC20B

Available on -20 variant only



B

C

D

VCC3V3

BANK 34

IO_0_34	R19	HDMI TX HPDN
IO_L1P_T0_34	T11	JB2 P
IO_L1N_T0_34	T10	JB2 N
IO_L2P_T0_34	T12	CK MOSI
IO_L2N_T0_34	U12	CK IO1
IO_L3P_T0_DQS_PUDC_B_34	U13	CK IO2
IO_L3N_T0_DQS_34	V13	CK IO3
IO_L4P_T0_34	V12	JB4 P
IO_L4N_T0_34	W13	JB4 N
IO_L5P_T0_34	T14	CK IO0
IO_L5N_T0_34	T15	CK IO5
IO_L6P_T0_34	P14	LED1
IO_L6N_T0_VREF_34	R14	LED0
IO_L7P_T1_34	Y16	JA2 P
IO_L7N_T1_34	Y17	JA2 N
IO_L8P_T1_34	W14	JB1 P
IO_L8N_T1_34	Y14	JB1 N
IO_L9P_T1_DQS_34	T16	CK IO10
IO_L9N_T1_DQS_34	U17	CK IO7
IO_L10P_T1_34	V15	CK IO4
IO_L10N_T1_34	W15	CK MISO
IO_L11P_T1_SRCC_34	U14	HDMI RX SCL
IO_L11N_T1_SRCC_34	U15	HDMI RX SDA
IO_L12P_T1_MRCC_34	U18	JA3 P
IO_L12N_T1_MRCC_34	U19	JA3 N
IO_L13P_T2_MRCC_34	N18	HDMI RX CLK P
IO_L13N_T2_MRCC_34	P19	HDMI RX CLK N
IO_L14P_T2_SRCC_34	N20	HDMI RX D2 P
IO_L14N_T2_SRCC_34	P20	HDMI RX D2 N
IO_L15P_T2_DQS_34	T20	HDMI RX D1 P
IO_L15N_T2_DQS_34	U20	HDMI RX D1 N
IO_L16P_T2_34	V20	HDMI RX D0 P
IO_L16N_T2_34	W20	HDMI RX D0 N
IO_L17P_T2_34	Y18	JA1 P
IO_L17N_T2_34	Y19	JA1 N
IO_L18P_T2_34	V16	JB3 P
IO_L18N_T2_34	W16	JB3 N
IO_L19P_T3_34	R16	CK IO6
IO_L19N_T3_VREF_34	R17	CK IO11
IO_L20P_T3_34	T17	AUD SD
IO_L20N_T3_34	R18	AUD PWM
IO_L21P_T3_DQS_34	V17	CK IO8
IO_L21N_T3_DQS_34	V18	CK IO9
IO_L22P_T3_34	W18	JA4 P
IO_L22N_T3_34	W19	JA4 N
IO_L23P_T3_34	N17	CK IO13
IO_L23N_T3_34	P18	CK IO12
IO_L24P_T3_34	P15	CK SDA
IO_L24N_T3_34	P16	CK SCL
IO_25_34	T19	HDMI RX HPD

IC20C

VCC3V3

R301
4.7K

CK_IO2

VCC3V3

BANK 35

IO_0_35	G14	LED5_B
IO_L1P_T0_AD0P_35	C20	AD0_P
IO_L1N_T0_AD0N_35	B20	AD0_N
IO_L2P_T0_AD8P_35	B19	AD8_P
IO_L2N_T0_AD8N_35	A20	AD8_N
IO_L3P_T0_DQS_AD1P_35	E17	CK AN0 P
IO_L3N_T0_DQS_AD1N_35	D18	CK AN0 N
IO_L4P_T0_35	D19	BTN0
IO_L4N_T0_35	D20	BTN1
IO_L5P_T0_AD9P_35	E18	CK AN1 P
IO_L5N_T0_AD9N_35	E19	CK AN1 N
IO_L6P_T0_35	F16	CK SS
IO_L6N_T0_VREF_35	F17	
IO_L7P_T1_AD2P_35	M19	SW1
IO_L7N_T1_AD2N_35	M20	SW0
IO_L8P_T1_AD10P_35	M17	HDMI TX SCL
IO_L8N_T1_AD10N_35	M18	HDMI TX SDA
IO_L9P_T1_DQS_AD3P_35	L19	BTN3
IO_L9N_T1_DQS_AD3N_35	L20	BTN2
IO_L10P_T1_AD11P_35	K19	HDMI TX D1 P
IO_L10N_T1_AD11N_35	J19	HDMI TX D1 N
IO_L11P_T1_SRCC_35	L16	HDMI TX CLK P
IO_L11N_T1_SRCC_35	L17	HDMI TX CLK N
IO_L12P_T1_MRCC_35	K17	HDMI TX D0 P
IO_L12N_T1_MRCC_35	K18	HDMI TX D0 N
IO_L13P_T2_MRCC_35	H16	SYSCLK
IO_L13N_T2_MRCC_35	H17	HDMI RX CEC
IO_L14P_T2_AD4P_SRCC_35	J18	HDMI TX D2 P
IO_L14N_T2_AD4N_SRCC_35	H18	HDMI TX D2 N
IO_L15P_T2_DQS_AD12P_35	F19	AD12 P
IO_L15N_T2_DQS_AD12N_35	F20	AD12 N
IO_L16P_T2_35	G17	LED4_G
IO_L16N_T2_35	G18	
IO_L17P_T2_AD5P_35	J20	CK AN4 P
IO_L17N_T2_AD5N_35	H20	CK AN4 N
IO_L18P_T2_AD13P_35	G19	CK AN5 P
IO_L18N_T2_AD13N_35	G20	CK AN5 N
IO_L19P_T3_35	H15	CK SCK
IO_L19N_T3_VREF_35	G15	HDMI TX CEC
IO_L20P_T3_AD6P_35	K14	CK AN2 P
IO_L20N_T3_AD6N_35	J14	CK AN2 N
IO_L21P_T3_DQS_AD14P_35	N15	LED4_R
IO_L21N_T3_DQS_AD14N_35	N16	LED2
IO_L22P_T3_AD7P_35	L14	LED5_G
IO_L22N_T3_AD7N_35	L15	LED4_B
IO_L23P_T3_35	M14	LED3
IO_L23N_T3_35	M15	LED5_R
IO_L24P_T3_AD15P_35	K16	CK AN3 P
IO_L24N_T3_AD15N_35	J16	CK AN3 N
IO_25_35	I15	CRYPTO_SDA

IC20D

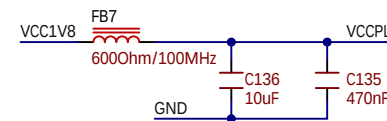
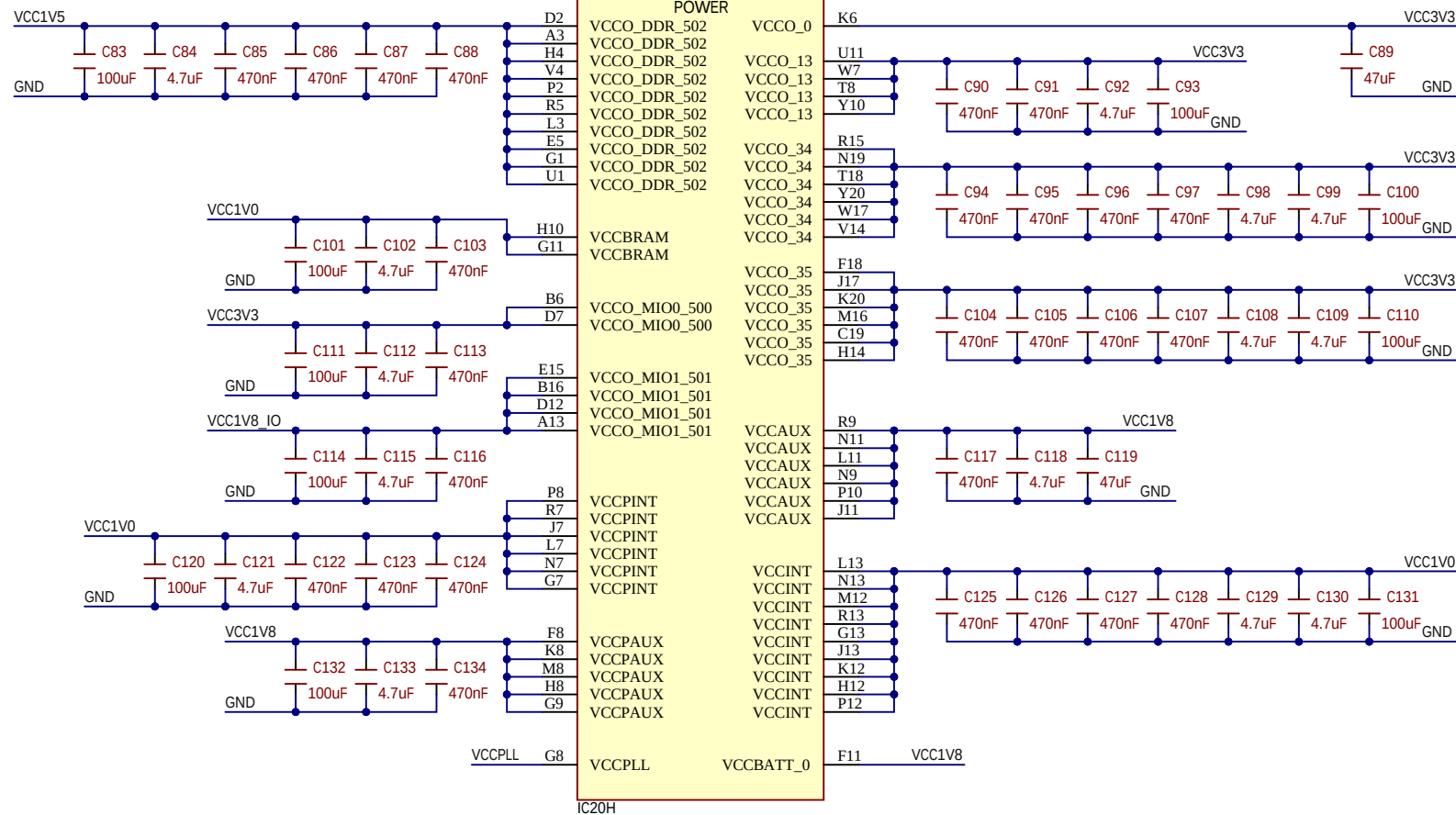
Title

Arty Z7

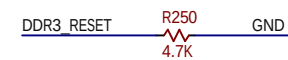
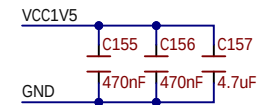
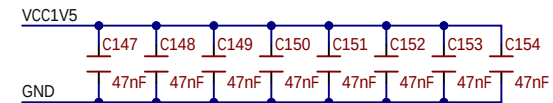
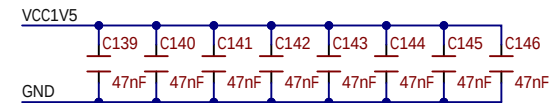
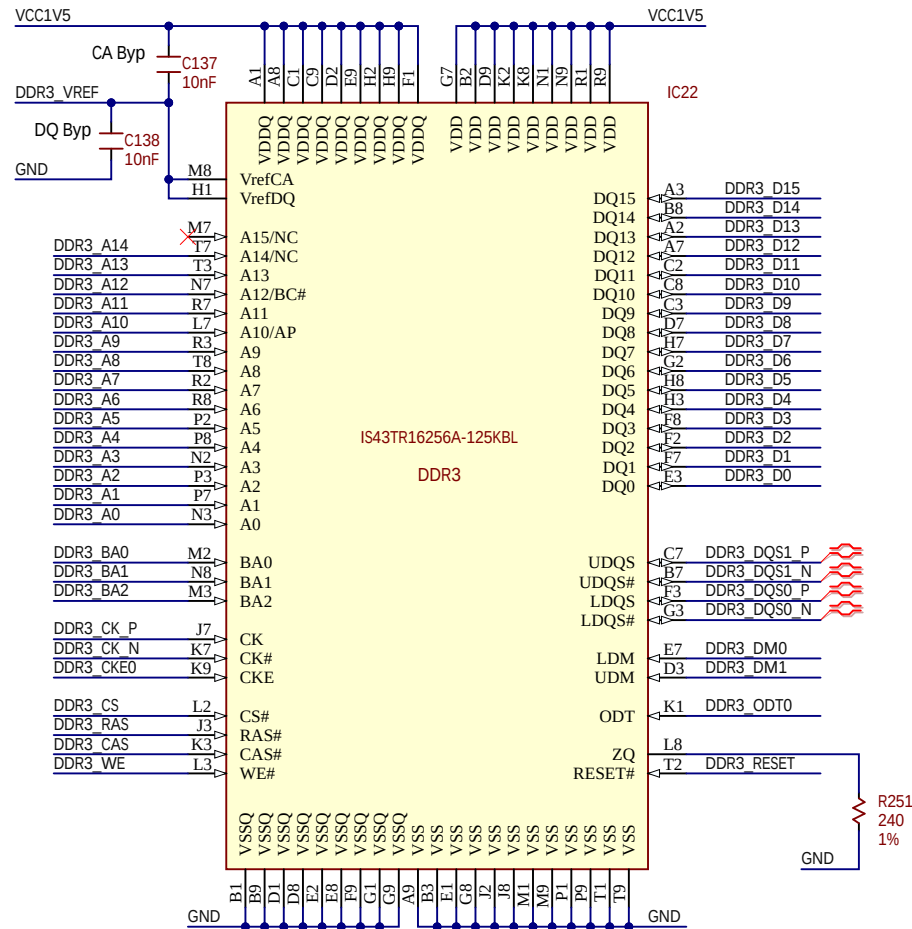
Rev

D.0

Circuit	
FPGA Banks, Anti Alias Filt.	
Doc#	500-346
Engineer	IC
Author	GMA
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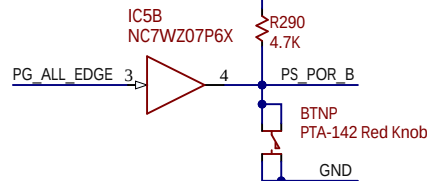
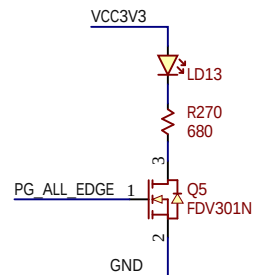
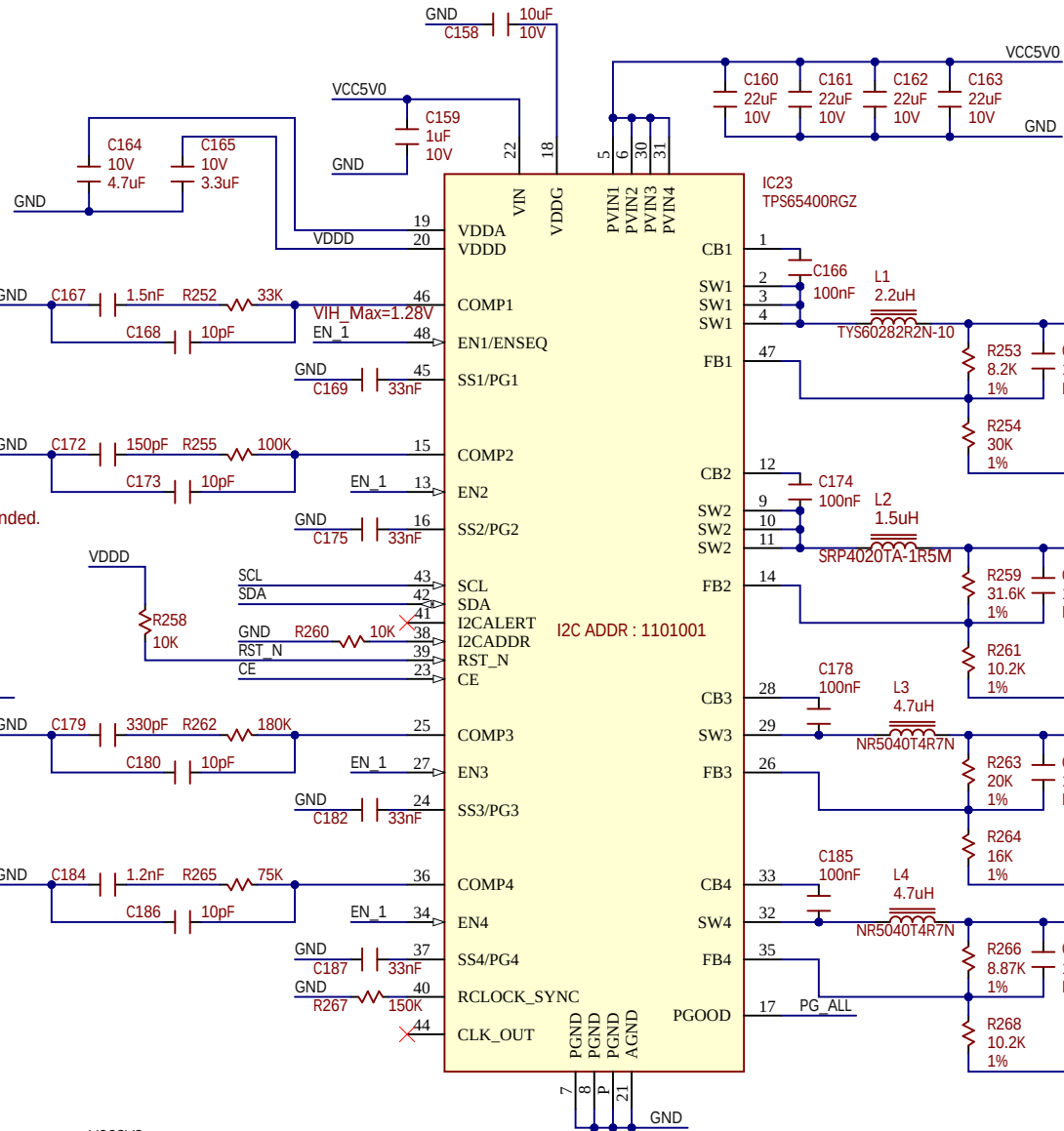
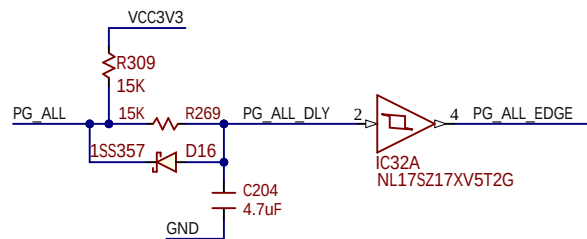
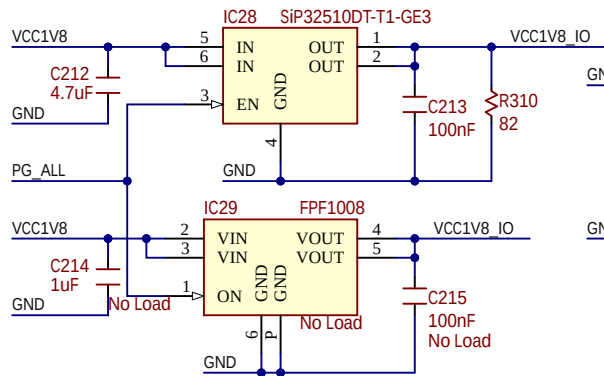
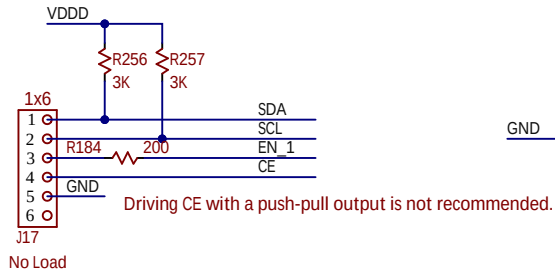


Title		Rev
Arty Z7		D.0
Circuit		
FPGA Power		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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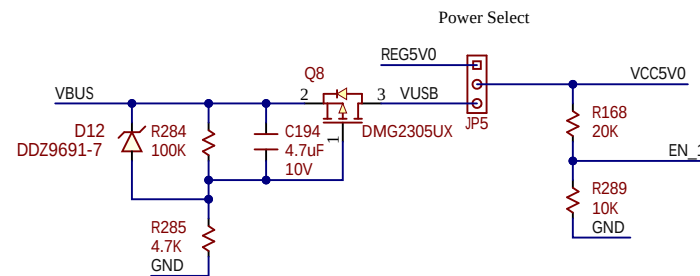
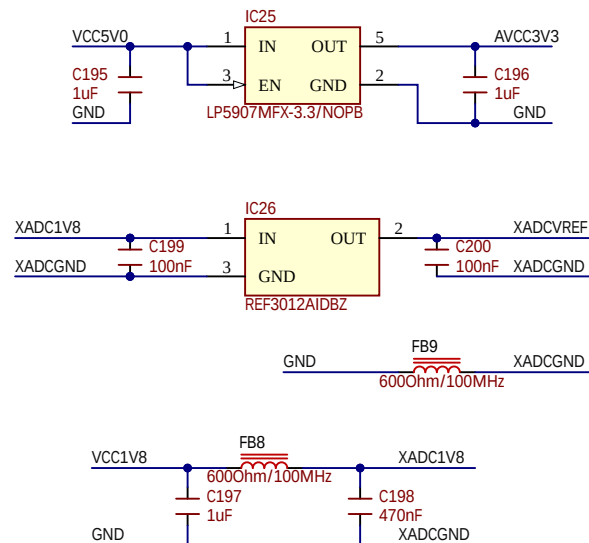


Title Arty Z7		Rev D.0
Circuit DDR3 Memory		
Doc# 500-346		
Engineer IC		
Author GMA		
Date 6/29/2022		
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1. VCC1V0
2. VCC1V8
3. VCC1V5
4. VCC3V3
5. VCC1V8_IO



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Power Regulation			
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